

Project Proposal

1. Proposed Project Title

KodeX

KodeX is a competitive programming platform enabling **problem setters to create problems with hidden and visible test cases**, and **users to solve problems with server-side code execution via Judge0 API**. Both user solution and tester's expected solution are executed against all test cases, automatically compared for exact output matching, and verdicts (AC/WA/TLE/MLE) displayed with detailed per-test-case breakdown. Built with MERN stack, it supports secure code execution, leaderboards.

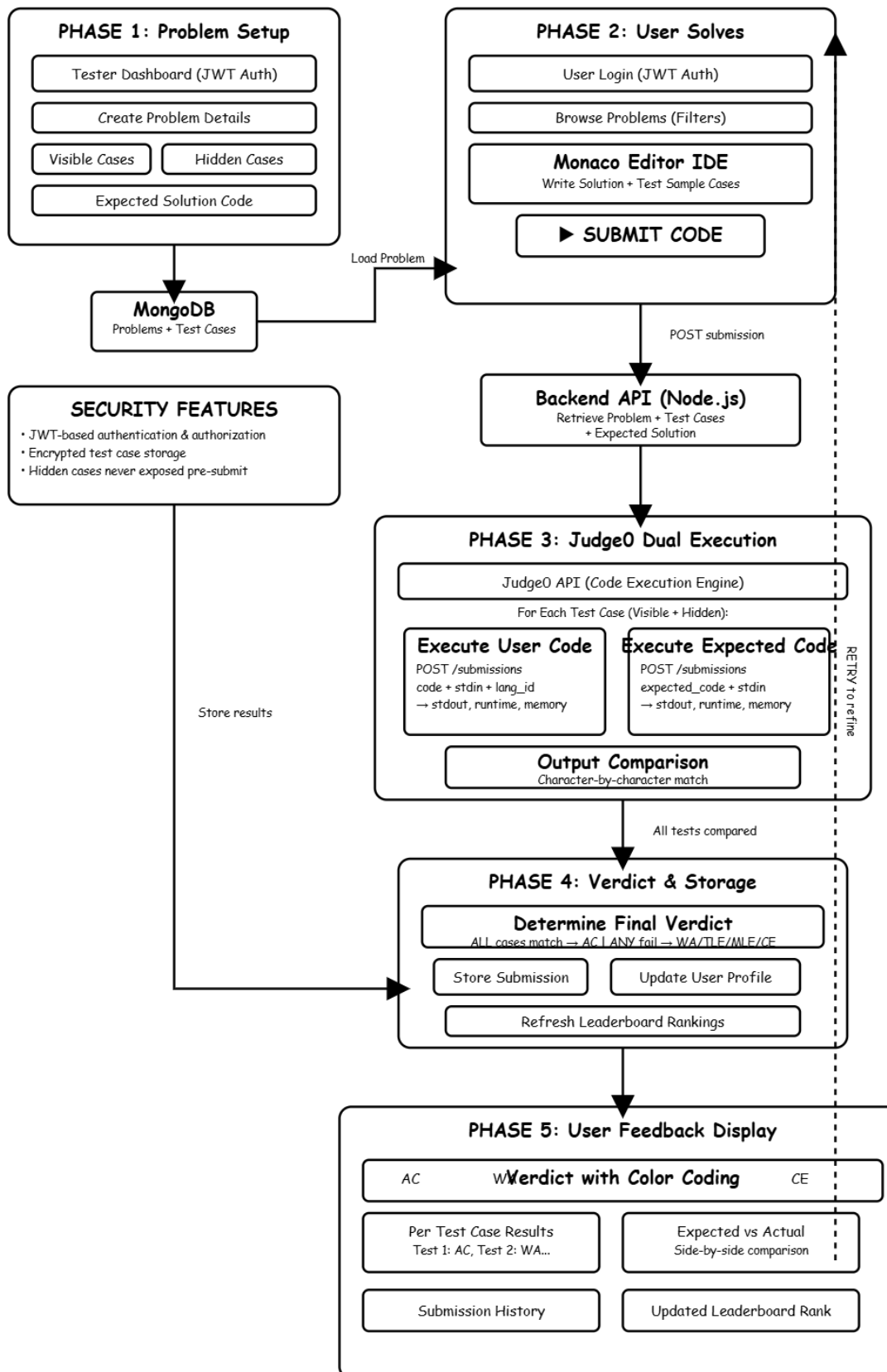
2. List of Functionalities

Core Features:

1. **Problem Setter Dashboard** – Testers create problems with title, difficulty, tags, description, sample test cases (visible), hidden test cases (input/output pairs), and expected solution code. Form validation ensures test case integrity before publication.
2. **User Authentication & Authorization** – JWT-based secure login/signup with role-based access control (student user, problem setter/tester, admin). Passwords hashed with bcrypt.
3. **Problem Repository Browser** – Users browse problems filtered by difficulty (Easy/Medium/Hard), tags (Arrays, Trees, DP, Graphs), and company-specific curated lists. Sample test cases displayed; hidden cases remain concealed.
4. **Online IDE with Monaco Editor** – Browser-based code editor supporting JavaScript, Java, Python, C++ with syntax highlighting, auto-indentation, real-time formatting, and code templates.
5. **Dual Code Execution Engine** – Backend simultaneously executes user solution and tester's expected solution against all test cases (visible + hidden) via Judge0 API. Outputs compared character-by-character for exact matches.

6. **Automated Verdict System** – Display verdicts (AC/WA/TLE/MLE/CE) with detailed metrics: runtime, memory usage, CPU time. Show per-test-case breakdown indicating which cases passed/failed and why (WA → show user vs expected output).
7. **Submission History & Analytics** – Track all submission attempts with code, timestamp, verdict, language, runtime/memory. Users view acceptance rate, total attempts per problem, and time-series of improvement.
8. **Global & Friend Leaderboards** – Real-time rankings by problems solved, acceptance rate, contest performance. Points system: easy +1, medium +2, hard +5. Weekly reset for competitive streaks.
9. **Community Discussion Forum** – Threaded comments per problem for solution hints, approach discussion, and Q&A. Users upvote helpful comments; tester can pin official solutions.
10. **Admin Dashboard** – Add/edit/delete problems, manage test cases, monitor platform analytics (submissions/day, average verdict time), moderate discussions, view server logs, manage user roles.

3. Block Diagram



4. Technologies to be Used

Frontend Technologies

- React.js 18+
- Tailwind CSS
- Monaco Editor
- Axios
- Redux Toolkit
- React Router v6

Backend Technologies

- Node.js 18+
- Express.js 4
- Judge0 CE API
- JWT (jsonwebtoken)
- Bcrypt
- Axios
- Dotenv
- CORS
- Helmet

Database & Infrastructure

- MongoDB
- Mongoose 7+
- MongoDB Atlas
- GitHub

Development & Testing Tools

- VS Code
- Postman / Thunder Client
- Git + GitHub
- ESLint + Prettier